CLASS implementation note: scalar-field couplings to matter and dark radiation

Branch `DE_DR_DM_int`

1 Purpose and conventions

This note summarizes the CLASS implementation of scalar-field couplings to matter and dark radiation, following the conventions of `REFS_FOR_CLAUDE/NotesRadiationCoupling260616.tex`. The paper example for the matter-coupled model is `2505.10410v2.pdf`; the two radiation-coupled models are those of `NotesRadiationCoupling_DONTTOUCH.pdf`.

We use a scalar field ϕ with potential $V(\phi)$ and multiplicative coupling functions $A_m(\phi)$ and $A_r(\phi)$ such that

$$\rho_m = A_m(\phi)\,\bar\rho_m, \qquad \rho_r = A_r(\phi)\,\bar\rho_r, \qquad \bar\rho_m \propto a^{-3}, \qquad \bar\rho_r \propto a^{-4}. \tag{1.1}$$

The fiducial background values used in the comparison runs are

$$\Omega_{r0} = 10^{-4}, \qquad \Omega_{m0} = 0.3149, \qquad \Omega_{\phi 0} = 0.6850, \qquad \Omega_{k0} = 0. \tag{1.2}$$

All implementation changes are inside CLASS. The external scripts used during development only read CLASS output tables and produced the figures included below.

2 Matter-coupled scalar background

For the matter sector, the new form is

$$A_m(\phi) = 1 + C_m\left(1 - e^{-\beta_m\phi}\right), \qquad C(\phi) = A_m(\phi)^2. \tag{2.1}$$

In the code this is selected by

$$\texttt{scf\_conformal\_form = matter\_exp}. \tag{2.2}$$

The existing qcdm–scalar conformal machinery is then reused through `C_scf`, `dC_scf`, `ddC_scf`, `Q_scf`, and the existing perturbed `delta_Q_scf` coefficients.

The option

$$\texttt{scf\_ic\_from\_today = yes} \tag{2.3}$$

reconstructs $\phi_{\rm ini}$, $\phi'_{\rm ini}$, V_0, and the initial qcdm density from the requested today values. This is used by `fig3_2505_matter.ini`.

The Fig. 3 reproduction uses exactly the parameter values of Eq. (3.8) of `2505.10410v2.pdf`. With the identification $C_m \equiv A_{m\varphi}$ and $\beta_m \equiv \alpha_m$ between Eq. (2.1) and Eq. (3.7) of the paper,

$$\lambda = \sqrt{2}, \qquad C_m = \frac{1}{8}, \qquad \beta_m = \sqrt{\tfrac{2}{3}}, \qquad w_{\varphi 0} = -0.76365999931, \tag{2.4}$$

set in the code as `scf_lambda = 1.4142135623730951`, `scf_C0 = 0.125`, `scf_beta = 0.816496580927726`, and `scf_w_phi_today = -0.76365999931`. (The slope is $\sqrt{2}$, not 2.)

For this case the effective dark-energy variables are

$$\rho_{\rm DE} = \rho_\phi + (A_m - 1)\bar\rho_m, \qquad w_{\rm DE}\,\rho_{\rm DE} = p_\phi. \tag{2.5}$$

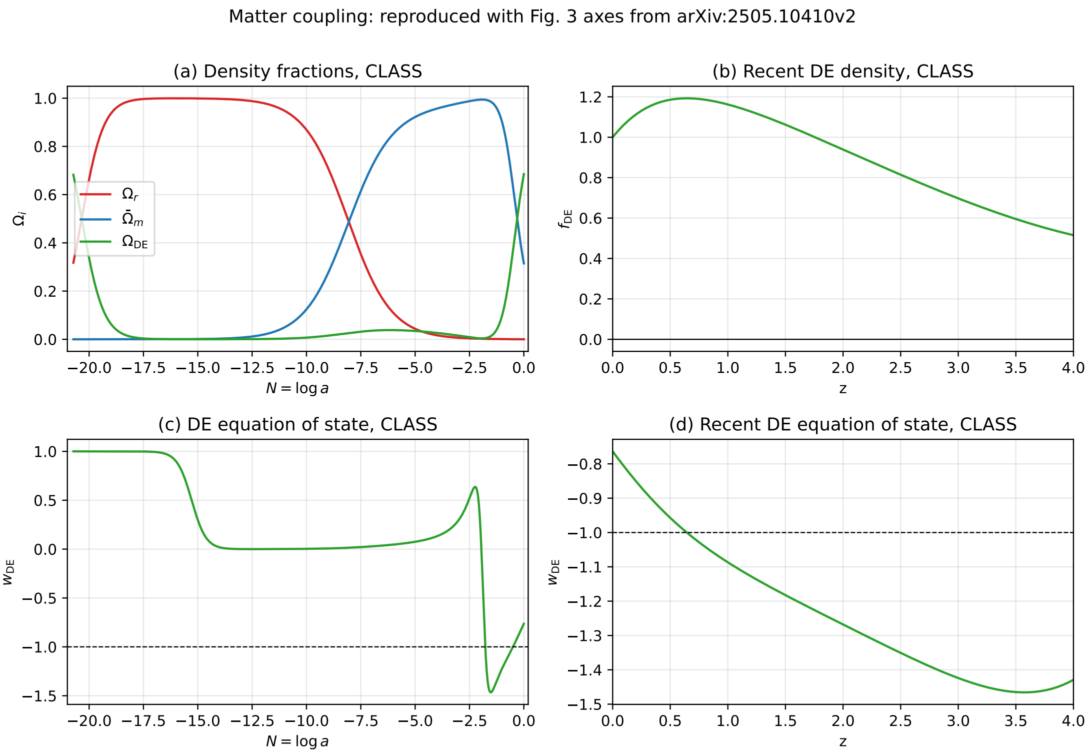

Figure 1: Matter-coupled CLASS reproduction for the `fig3_2505_matter.ini` setup. The plotted matter fraction uses $\bar{\rho}_m = \rho_{\rm qcdm}/A_m$.

At $z = 0$, the CLASS output gives

quantity	CLASS value
$w_{\phi 0}$	-0.763660007
$w_{\rm DE0}$	-0.763660007
Ω_{r0}	9.9999997×10^{-5}
Ω_{m0}	0.314900009
$\Omega_{\rm DE0}$	0.684999991
ϕ_0	-1.31×10^{-8}

3 Radiation-coupled scalar background

The radiation coupling is independent of the qcdm conformal factor. It is selected by

$$\texttt{scf\_interacting\_radiation = yes} \tag{3.1}$$

and uses the free parameters `scf_C0_r` and `scf_beta_r`:

$$A_r(\phi) = 1 + C_r \left(1 - e^{-\beta_r \phi}\right). \tag{3.2}$$

The CLASS background density stored in `rho_idr` is the full coupled density,

$$\rho_{\rm idr} = A_r(\phi)\, \bar{\rho}_{\rm idr}. \tag{3.3}$$

The pressure is kept as radiation,

$$p_{\rm idr} = \frac{1}{3}\rho_{\rm idr}. \tag{3.4}$$

The normalization of the radiation Lagrangian only fixes the field amplitude inside $\bar{\rho}_{\rm idr}$. The implemented object is the radiation stress tensor,

$$T^0{}_0 = -\rho_{\rm idr}, \qquad T^i{}_j = \frac{1}{3}\rho_{\rm idr}\delta^i_j, \tag{3.5}$$

so A_r rescales the full dark-radiation tensor. The anisotropic-stress hierarchy is otherwise the existing CLASS idr hierarchy.

The scalar background equation is implemented in CLASS density units as

$$\phi'' + 2\mathcal{H}\phi' + a^2\left(V_{,\phi} - Q_m - Q_r\right) = 0, \qquad Q_r = -3\rho_{\rm idr}\,\frac{\partial_\phi A_r}{A_r}. \tag{3.6}$$

This sign matches the physical equation

$$\ddot{\phi} + 3H\dot{\phi} + V_{,\phi} + \bar{\rho}_m\partial_\phi A_m + \bar{\rho}_r\partial_\phi A_r = 0, \tag{3.7}$$

after using the CLASS convention in which densities are multiplied by $8\pi G/3$.

The effective dark-energy output is

$$\rho_{\rm DE} = \rho_\phi + (A_m - 1)\bar{\rho}_m + (A_r - 1)\bar{\rho}_r, \tag{3.8}$$

$$w_{\rm DE}\,\rho_{\rm DE} = p_\phi + \frac{1}{3}(A_r - 1)\bar{\rho}_r. \tag{3.9}$$

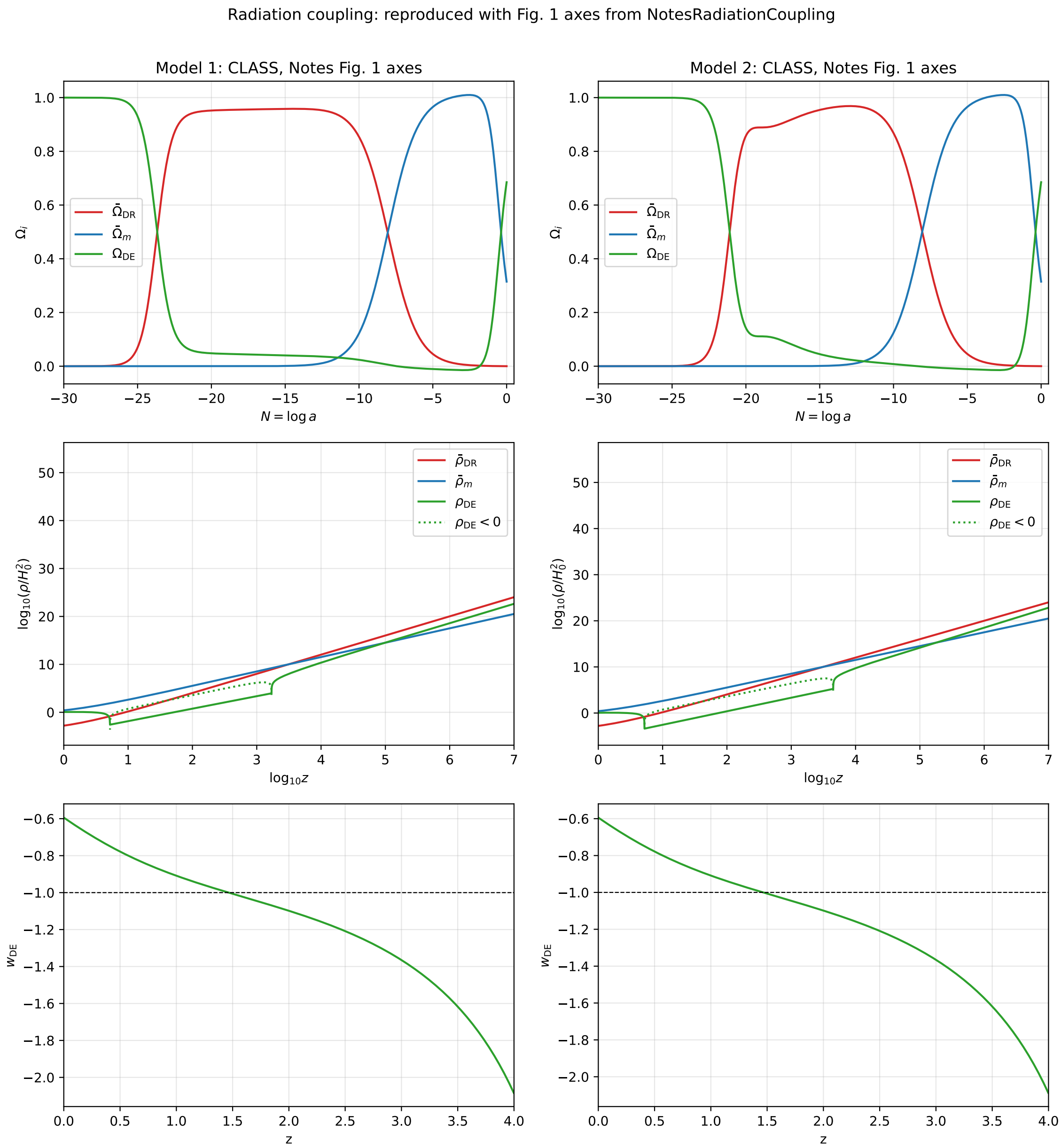

Figure 2: Radiation-coupled CLASS reproduction for the two `NotesRadiationCoupling` Fig. 1 models. The plotted barred radiation and matter densities are reconstructed as $\bar{\rho}_r = \rho_{\rm idr}/A_r$ and $\bar{\rho}_m = \rho_{\rm qcdm}/A_m$. The middle row shows the effective dark-energy density $\rho_{\rm DE}$ [Eq. (3.8)]; since $\rho_{\rm DE}$ can change sign where $(A_r - 1) < 0$, we plot $\log_{10} |\rho_{\rm DE}/H_0^2|$ and draw the negative stretches with a dotted line.

At $z = 0$, the two Notes models give

quantity	model 1	model 2
$w_{\phi 0}$	-0.594208876	-0.594223632
$w_{\rm DE0}$	-0.594208863	-0.594223633
Ω_{r0}	1.0000022×10^{-4}	9.9999990×10^{-5}
Ω_{m0}	0.314899728	0.314900018
$\Omega_{\rm DE0}$	0.685000288	0.684999981
ϕ_0	1.30×10^{-6}	-5.69×10^{-8}
$A_r(\phi_0)$	0.999999901	1.000000001

4 Radiation perturbations

The perturbations are implemented for the full coupled idr variables

$$\delta_{\rm idr} \equiv \frac{\delta\rho_{\rm idr}}{\rho_{\rm idr}}, \qquad \theta_{\rm idr}. \tag{4.1}$$

Define

$$\Lambda_r \equiv \frac{\partial_\phi A_r}{A_r}, \qquad \Lambda_{2,r} \equiv \frac{{\rm d}^2 \ln A_r}{{\rm d}\phi^2} = \frac{\partial_\phi^2 A_r}{A_r} - \Lambda_r^2. \tag{4.2}$$

In synchronous gauge, the added idr terms are

$$\delta_{\rm idr}' = -\frac{4}{3}\left(\theta_{\rm idr} + \frac{h'}{2}\right) + \Lambda_r\,\delta\phi' + \Lambda_{2,r}\,\phi'\,\delta\phi, \tag{4.3}$$

$$\theta_{\rm idr}' = \frac{k^2}{4}\delta_{\rm idr} - k^2\sigma_{\rm idr} + \frac{3}{4}\Lambda_r k^2\delta\phi - \Lambda_r\phi'\theta_{\rm idr}. \tag{4.4}$$

IDM–DR scattering terms, when present, are preserved and the scalar force is added on top of the usual idr Euler equation.

The scalar perturbation equation receives

$$\delta Q_r = -3\rho_{\rm idr}\left(\Lambda_r\delta_{\rm idr} + \Lambda_{2,r}\delta\phi\right), \tag{4.5}$$

which enters the perturbed Klein–Gordon equation as $a^2\delta Q_r$, analogously to the existing qcdm `delta_Q_scf` term. In Newtonian gauge the metric term $2\psi Q_r$ is added in the same way as for the qcdm source.

5 Code changes: before and after

This section documents the concrete code edits behind the equations above, as *before/after* pairs taken from the three implementation commits (`11432659`, `2f4abc0d`, `dfb1975c`) relative to the pre-implementation state (`74a2ed72`). Line numbers refer to the current tree. Blocks marked "new" had no prior counterpart.

5.1 `source/background.c`: matter conformal form

The conformal factor selector gained the `matter_exp` branch $C(\phi) = A_m(\phi)^2$ with $A_m = 1+C_m(1-e^{-\beta_m\phi})$ [Eq. (2.1)]. Shown for `C_scf`; `dC_scf` and `ddC_scf` were extended identically.

`Before (C_scf, only the exponential form existed):`

```c
switch (pba->scf_conformal_form) {
  case scf_conformal_exp:
    return scf_C0*exp(2.*scf_beta*phi);
  default:
    class_stop(pba->error_message, "Unknown scf_conformal_form %d", pba->scf_conformal_form);
    return 1.0;
}
```

After (new matter_exp case, plus a shared helper):

```c
static void scf_matter_exp_A(struct background *pba, double phi,
                             double *A, double *dA, double *ddA) {
  double Aphi = pba->C0_scf, alpha = pba->beta_scf, e = exp(-alpha*phi);
  *A = 1.0 + Aphi*(1.0 - e);
  *dA = Aphi*alpha*e;
  *ddA = -Aphi*alpha*alpha*e;
}
/* ... inside C_scf: */
switch (pba->scf_conformal_form) {
  case scf_conformal_exp:
    return scf_C0*exp(2.*scf_beta*phi);
  case scf_conformal_matter_exp: /* NEW */
    scf_matter_exp_A(pba,phi,&A,&dA,&ddA);
    return A*A;
  default:
    class_stop(pba->error_message, "Unknown scf_conformal_form %d", pba->scf_conformal_form);
    return 1.0;
}
```

5.2 source/background.c: radiation coupling function (new)

The multiplicative radiation factor A_r and its derivatives [Eq. (3.2)] were added with the same exponential template:

New (source/background.c, ~l. 3661--3715):

```c
static void scf_radiation_exp_A(struct background *pba, double phi,
                                double *A, double *dA, double *ddA) {
  double Aphi = pba->C0_r_scf, beta = pba->beta_r_scf, e = exp(-beta*phi);
  *A = 1.0 + Aphi*(1.0 - e);
  *dA = Aphi*beta*e;
  *ddA = -Aphi*beta*beta*e;
}
double A_r_scf(...) { ... return A; } /* A_r(phi) */
double dA_r_scf(...) { ... return dA; } /* d A_r / dphi */
double ddA_r_scf(...){ ... return ddA;} /* d^2 A_r / dphi^2 */
```

5.3 source/background.c: coupled rho_idr

The interacting-dark-radiation density is multiplied by $A_r(\phi)$ so that **rho_idr** stores the full coupled density [Eq. (3.3)].

Before (~l. 630):

```c
if (pba->has_idr == _TRUE_) {
  pvecback[pba->index_bg_rho_idr] = pba->Omega0_idr * pow(pba->H0,2) / pow(a,4);
  rho_tot += pvecback[pba->index_bg_rho_idr];
  ...
}
```

After:

```c
if (pba->has_idr == _TRUE_) {
  pvecback[pba->index_bg_rho_idr] = pba->Omega0_idr * pow(pba->H0,2) / pow(a,4);
  if (pba->has_scf_radiation == _TRUE_) { /* NEW */
    pvecback[pba->index_bg_rho_idr] *= A_r_scf(pba,phi); /* NEW */
  }
  rho_tot += pvecback[pba->index_bg_rho_idr];
  ...
}
```

5.4 source/background.c: scalar background equation

The background Klein–Gordon source $-Q$ in $\phi''+2\mathcal{H}\phi'+a^2(V_{,\phi}-Q)=0$ gained the radiation term $Q_r = -3\rho_{\rm idr}\,\partial_\phi A_r/A_r$ [Eq. (3.6)].

Before (~l. 3105):

```c
if (pba->has_scf == _TRUE_) {
  double Q_bg_scf = 0.;
  if (pba->has_qcdm_de_q == _TRUE_) {
    Q_bg_scf = Q_scf(pba, y[..phi], y[..phi_prime], y[..rho_qcdm], a, pvecback);
  }
  dy[pba->index_bi_phi_scf] = y[pba->index_bi_phi_prime_scf]/a/H;
  ...
}
```

After:

```c
if (pba->has_scf == _TRUE_) {
  double Q_bg_scf = 0.;
  if (pba->has_qcdm_de_q == _TRUE_) {
    Q_bg_scf = Q_scf(pba, y[..phi], y[..phi_prime], y[..rho_qcdm], a, pvecback);
  }
  if (pba->has_scf_radiation == _TRUE_) { /* NEW */
    double A_r = A_r_scf(pba, y[pba->index_bi_phi_scf]);
    class_test(A_r <= 0., error_message, "A_r(phi) = %e must stay positive.", A_r);
    /* rho_idr stores A_r rho_bar_r; CLASS units give Q_r = -3 rho_bar_r A_r,phi */
    Q_bg_scf += -3.0*(pvecback[pba->index_bg_rho_idr]/A_r)*dA_r_scf(pba, y[pba->index_bi_phi_scf]);
  }
  dy[pba->index_bi_phi_scf] = y[pba->index_bi_phi_prime_scf]/a/H;
  ...
}
```

5.5 source/background.c: effective dark-energy output (new)

The output block was extended to print $\rho_{\rm DE}$ and $w_{\rm DE}$ including the radiation term [Eqs. (3.8)–(3.9)]:

New (~l. 2924):

```c
double A_r = (pba->has_scf_radiation==_TRUE_) ? pvecback[pba->index_bg_A_r_scf] : 1.0;
double rho_bar_r = (pba->has_scf_radiation==_TRUE_) ? pvecback[pba->index_bg_rho_idr]/A_r : 0.0;
double matter_coupling = 0.0;
if (pba->has_qcdm_de_q == _TRUE_) {
  double A_scf = sqrt(pvecback[pba->index_bg_C_scf]); /* A_m = sqrt(C) */
  matter_coupling = pvecback[pba->index_bg_rho_qcdm]*(1.0 - 1.0/A_scf);
}
double rho_de_eff = pvecback[pba->index_bg_rho_scf] + matter_coupling + (A_r-1.0)*rho_bar_r;
double p_de_eff = pvecback[pba->index_bg_p_scf] + (1.0/3.0)*(A_r-1.0)*rho_bar_r;
```

5.6 source/perturbations.c: coupling coefficients (new)

Λ_r, $\Lambda_{2,r}$ and Q_r [Eq. (4.2)] are precomputed once per call:

New (~l. 9365):

```
if (use_radiation == _TRUE_) {
  lambda_r_scf = pvecback[pba->index_bg_dA_r_scf]/pvecback[pba->index_bg_A_r_scf];
  lambda2_r_scf = pvecback[pba->index_bg_ddA_r_scf]/pvecback[pba->index_bg_A_r_scf]
                  - lambda_r_scf*lambda_r_scf;
  Q_r_scf = -3.0*pvecback[pba->index_bg_rho_idr]*lambda_r_scf;
}
```

5.7 source/perturbations.c: idr continuity and Euler

The two source terms of Eqs. (4.3)–(4.4) are added on top of the standard idr fluid equations.

Before (~l. 9979):

```
dy[pv->index_pt_delta_idr] = -4./3.*(theta_idr + metric_continuity);
...
dy[pv->index_pt_theta_idr] = k2*(y[pv->index_pt_delta_idr]/4.-s2_squared*y[pv->index_pt_shear_idr]) +
     metric_euler;
```

After:

```
dy[pv->index_pt_delta_idr] = -4./3.*(theta_idr + metric_continuity);
if (use_radiation == _TRUE_) { /* NEW */
  dy[pv->index_pt_delta_idr] +=
      lambda_r_scf*y[pv->index_pt_phi_prime_scf]
    + lambda2_r_scf*pvecback[pba->index_bg_phi_prime_scf]*y[pv->index_pt_phi_scf];
}
...
dy[pv->index_pt_theta_idr] = k2*(y[pv->index_pt_delta_idr]/4.-s2_squared*y[pv->index_pt_shear_idr]) +
     metric_euler;
if (use_radiation == _TRUE_) { /* NEW */
  dy[pv->index_pt_theta_idr] +=
      0.75*lambda_r_scf*k2*y[pv->index_pt_phi_scf]
    - lambda_r_scf*pvecback[pba->index_bg_phi_prime_scf]*theta_idr;
}
```

5.8 source/perturbations.c: perturbed Klein–Gordon source

The radiation source δQ_r [Eq. (4.5)] is built and injected into the perturbed KG equation as $a^2\delta Q_r$, with the Newtonian-gauge $2\psi Q_r$ term, mirroring the qcdm **delta_Q_scf** source.

Before (synchronous KG, no radiation term):

```
dy[pv->index_pt_phi_prime_scf] =
    -2.*a_prime_over_a*y[pv->index_pt_phi_prime_scf]
    -(k2+a2*pvecback[pba->index_bg_ddV_scf])*y[pv->index_pt_phi_scf]
    -metric_continuity*pvecback[pba->index_bg_phi_prime_scf]
    + kg_term_entropy + kg_term_q;
```

After (source assembled, then added):

```
if (use_radiation == _TRUE_) { /* NEW */
  delta_Q_r_scf = Q_r_scf*delta_idr
                - 3.0*pvecback[pba->index_bg_rho_idr]*lambda2_r_scf*y[pv->index_pt_phi_scf];
}
...
double kg_term_r = 0.;
```

```
if (use_radiation == _TRUE_) { /* NEW */
  kg_term_r = a2*delta_Q_r_scf;
  if (ppt->gauge == newtonian)
    kg_term_r += 2.*a2*pvecmetric[ppw->index_mt_psi]*Q_r_scf;
}
dy[pv->index_pt_phi_prime_scf] =
    -2.*a_prime_over_a*y[pv->index_pt_phi_prime_scf]
    -(k2+a2*pvecback[pba->index_bg_ddV_scf])*y[pv->index_pt_phi_scf]
    -metric_continuity*pvecback[pba->index_bg_phi_prime_scf]
    + kg_term_entropy + kg_term_q + kg_term_r; /* + kg_term_r NEW */
```

5.9 `source/input.c` and `include/background.h`

Parsing of the new switches and storage of the new parameters/indices:

- `include/background.h`: added flags `has_scf_radiation`, parameters `C0_r_scf`, `beta_r_scf`, the enum value `scf_conformal_matter_exp`, and the background indices `index_bg_A_r_scf`, `index_bg_dA_r_scf`, `index_bg_ddA_r_scf`.
- `source/input.c`: parsing of `scf_interacting_radiation`, `scf_C0_r`, `scf_beta_r`, and the `scf_conformal_form = matter_exp` keyword, together with the `scf_ic_from_today` reconstruction used by the Fig. 3 setup.

6 Implemented files and validation

The implementation is distributed over the following files:

- `include/background.h`: new radiation-coupling parameters and background indices.
- `source/input.c`: parsing of `scf_interacting_radiation`, `scf_C0_r`, and `scf_beta_r`.
- `source/background.c`: A_r, $\partial_\phi A_r$, $\partial_\phi^2 A_r$, coupled `rho_idr`, scalar background source, and effective dark-energy outputs.
- `source/perturbations.c`: idr continuity/Euler source terms and the radiation contribution to the perturbed scalar equation.
- `fig3_2505_matter.ini`, `notes_fig1_model1.ini`, and `notes_fig1_model2.ini`: reproduction configurations.

The following checks were run:

- `make class`
- `./class fig3_2505_matter.ini`
- `./class notes_fig1_model1.ini`
- `./class notes_fig1_model2.ini`

The perturbation code was additionally smoke-tested with a temporary default-like cosmology containing a small scalar-field density and a small idr density, using `output = mPk`. That run completed the background, thermodynamics, source, primordial, and Fourier modules and wrote a matter power spectrum.

The two comparison figures are regenerated directly from the CLASS background tables by the self-contained script `doc/figures/make_comparison_figs.py`, which reads `output/fig3_2505_matter_b` `output/notes_fig1_model1_background.dat`, and `output/notes_fig1_model2_background.dat` and writes the `.pdf` and `.svg` versions of both figures. Rerunning `make class`, the three `.ini` files above and that script reproduces the $z = 0$ tables and the figures shown here.

7 Comments and limitations

The exact Notes Fig. 1 configurations are designed for background comparisons: they use `Omega_g = 1.e-12` and a very large `xi_idr` so photons are only a trace component while idr carries the desired radiation budget. This is not a healthy thermodynamics setup for CMB perturbation tests, so perturbative stability was tested on a default-like temporary configuration instead.

The current radiation-coupling function is the exponential form (3.2). The parameters are free, but adding other functional forms would require extending `A_r_scf`, `dA_r_scf`, and `ddA_r_scf` in `source/background.c`.